THE EULER CHARACTERISTIC OF A PERFECT COMPLEX IS WELL DEFINED ON THE DERIVED CATEGORY AND ADDITIVE ON DISTINGUISHED TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author's displayed chain of short exact sequences contains a misprint in the second-to-last line, which reads '0 -> N^{b-3} -> M^{b-2} -> N^{b-2} -> 0' where the pattern of the neighbouring lines suggests N^{b-2}; the line is reproduced unchanged rather than silently corrected.

Allowed standard prerequisites (editorial).

- Perfect objects of D(R) and their representation by finite complexes of finite projective modules. used at proof lines 36508-36509. Supplies the representatives on which the formula for c is defined.
- K_0(R) and additivity of the class map on split short exact sequences of finite projective modules. used at proof lines 36512 and 36529. Is the target group and the relation that makes both the surjection step and the telescoping computation work.
- Distinguished triangles of perfect complexes. condition (2) at statement lines 36499-36500. Fixes the additivity property that the constructed c must satisfy.

Source reference. Original lemma 014E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1469–1469. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 014Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1490–1490. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. There is a map*

$$c : \text{perfect complexes over } R \longrightarrow K_0(R)$$

with the following properties

(1) $c(K[n]) = (-1)^n c(K)$ *for a perfect complex* K,
(2) *if* $K \to L \to M \to K[1]$ *is a distinguished triangle of perfect complexes, then* $c(L) = c(K) + c(M)$,
(3) *if* K *is represented by a finite complex* $M^\bullet$ *consisting of finite projective modules, then* $c(K) = \sum(-1)^i[M_i]$.

Proof. Let K be a perfect object of $D(R)$. By definition we can represent K by a finite complex $M^\bullet$ of finite projective R-modules. We define c by setting

$$c(K) = \sum(-1)^n[M^n]$$

in $K_0(R)$. Of course we have to show that this is well defined, but once it is well defined, then (1) and (3) are immediate. For the moment we view the map c as defined on complexes of finite projective R-modules.

Suppose that $L^\bullet \to M^\bullet$ is a surjective map of finite complexes of finite projective R-modules. Let $K^\bullet$ be the kernel. Then we obtain short exact sequences of R-modules

$$0 \to K^n \to L^n \to M^n \to 0$$

which are split because M^n is projective. Hence $K^\bullet$ is also a finite complex of finite projective R-modules and $c(L^\bullet) = c(K^\bullet) + c(M^\bullet)$ in $K_0(R)$.

Suppose given finite complex $M^\bullet$ of finite projective R-modules which is acyclic. Say $M^n = 0$ for $n \notin [a, b]$. Then we can break $M^\bullet$ into short exact sequences

$$\begin{gathered} 0 \to M^a \to M^{a+1} \to N^{a+1} \to 0, \\ 0 \to N^{a+1} \to M^{a+2} \to N^{a+3} \to 0, \\ \ldots \\ 0 \to N^{b-3} \to M^{b-2} \to N^{b-2} \to 0, \\ 0 \to N^{b-2} \to M^{b-1} \to M^b \to 0 \end{gathered}$$

Arguing by descending induction we see that $N^{b-2}, \ldots, N^{a+1}$ are finite projective R-modules, the sequences are split exact, and

$$c(M^\bullet) = \sum(-1)[M^n] = \sum(-1)^n([N^{n-1}] + [N^n]) = 0$$

Thus our construction gives zero on acyclic complexes.

It follows formally from the results of the preceding two paragraphs that c is well defined and satisfies (2). Namely, suppose the finite complexes $M^\bullet$ and $L^\bullet$ of finite projective R-modules represent the same object of $D(R)$. Then we can represent the isomorphism by a map $f : M^\bullet \to L^\bullet$ of complexes, see Derived Categories, Lemma 064B. We obtain a short exact sequence of complexes

$$0 \to L^\bullet \to C(f)^\bullet \to K^\bullet[1] \to 0$$

see Derived Categories, Definition 014E. Since f is a quasi-isomorphism, the cone $C(f)^\bullet$ is acyclic (this follows for example from the discussion in Derived Categories, Section 014Z). Hence

$$0 = c(C(f)^\bullet) = c(L^\bullet) + c(K^\bullet[1]) = c(L^\bullet) - c(K^\bullet)$$

as desired. We omit the proof of (2) which is similar. □